

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1041

Westboro 100 kV
January 25, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1041. The Interconnection Customer proposes a 443.6 MW natural gas combined cycle facility interconnecting at the Westboro 100 kV substation in the Indiana area. The request was received on August 19, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 358 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$86,368,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LIMON WIND 2-BRANDFORD 100kV	SAPONY-WESTBORO 100kV	372.9	111	100.0
WESBORO-BRANDFORD 100kV	WESTBORO-SAPONY 100kV Line	356.6	122	100.0
WESTBORO SUBSTATION-SAPONY 100kV	WESTBORO-BRANDFORD 100kV L	342.9	127	100.0
LIMON WIND 2-MERIDIAN 100kV	WESTBORO-SAPONY 100kV line	340.9	117	100.0
STAGECOACH-SAPONY SUBSTATION 100kV	WESTBORO-BRANDFORD 100kV	299.5	141	100.0
BARRE-RAYMOND F. BARKER 100kV	BRANDFORD-WESTBORO 100kV L	249.3	157	100.0
MERIDIAN-ELTOPIA 100kV	WESTBORO-SAPONY 100kV	245.8	149	100.0
PORTAGE UPPCO-SORENSEN SUBSTATION 100kV	WESTBORO-SAPONY 100kV	188.3	186	100.0
BEDFORD RURAL-CHINA 100kV	WABASH 500-230 kV XFMR	187.7	108	10.4
RAYMOND F. BARKER-WABASH SUBSTATION 100kV	BRANDFORD-WESTBORO 100kV	185.6	201	100.0
HIGH SPRINGS-STAGECOACH 100kV	WESTBORO-BRANDFORD 100kV	183.3	121	53.3
CHINA-PONNEQUIN 100kV	WABASH 500-230 kV XFMR	183.0	119	10.4
SORENSEN SUBSTATION-ELTOPIA 100kV L	SAPONY-WESTBORO 100kV	176.3	206	100.0
BARRE-HIGH SPRINS 100kV	WESTBORO-BANDFORD 100kV	167.3	127	53.3
DEER CRK-PONNEQUIN 100kV Line	WABASH 500 kV/230 kV Autotransformer	157.9	152	10.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
INDEGO-WAYNEDALE SUBSTATION 100kV	MONFORT-WAYNEDALE 230kV	156.6	134	25.0
BARRE-STAGECOACH 100kV	WESTBORO-BRANDFORD 100kV line	156.5	122	46.7
WAYNEDALE SUBSTATION-AEP PETTIT AVENUE 100kV #1	MUSKOGEE-MONTPELIER ELECTRIC GENERATING STATION 500kV	134.3	220	6.9
WAYNEDALE-AEP PETTIT AVENUE 100kV	MUSKOGEE-MONTPELIER ELECTRIC GENERATING STA 500kV	134.3	220	6.9
INDEGO-COLUMBIA HILLS 100kV	WABASH 500-230 kV Autotransformer	131.6	194	26.0
STAUNTON-PIQUA POWER PLANT 230kV	E METHUEN 500/230kV XFMR	129.7	517	5.1
ILLINOIS ROAD-HEBRON ROSEMEADE 100kV	WAYNEDALE-INDEGO 100kV	129.5	109	26.0
SEGS 5-AEP PETTIT AVENUE 100kV	MUSKOGEE-MONTPELIER ELECTRIC GENERATING STA 500kV	129.3	178	6.6
CARLEY RD-PIQUA POWER LANT 230kV	E METHUEN 500/230kV Transformer	124.3	540	5.1
STAUNTON-BARAGA VLG 230kV	E METHUEN 500 kV/230 kV Transformer Bank	121.6	532	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LIMON WIND 2-BRANDFORD 100kV	Rebuild line with higher-capacity conductor	\$4,758,000	100% (\$4,758,000)
WESBORO-BRANDFORD 100kV	Rebuild line with higher-capacity conductor	\$4,616,000	100% (\$4,616,000)
WESTBORO SUBSTATION-SAPONY 100kV	Rebuild line with higher-capacity conductor	\$3,737,000	100% (\$3,737,000)
MERIDIAN-LIMON WIND 2 100kV Line	Rebuild line with higher-capacity conductor	\$3,907,000	100% (\$3,907,000)
SAPONY-STAGECOACH 100kV	Rebuild line with higher-capacity conductor	\$7,168,000	100% (\$7,168,000)
BARRE-RAYMOND F. BARKER 100kV	Rebuild line with higher-capacity conductor	\$8,061,000	100% (\$8,061,000)
MERIDIAN-ELTOPIA 100kV Line	Rebuild line with higher-capacity conductor	\$5,983,000	100% (\$5,983,000)
PORTAGE UPPCO-SORENSEN 100kV	Rebuild line with higher-capacity conductor	\$3,352,000	100% (\$3,352,000)
BEDFORD RURAL-CHINA 100kV	Rebuild line with higher-capacity conductor	\$4,754,000	53% (\$2,517,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RAYMOND F. BARKER-WABASH SUBSTATION 100kV	Rebuild line with higher-capacity conductor	\$5,927,000	100% (\$5,927,000)
HIGH SPRINGS-STAGECOACH 100kV	Rebuild line with higher-capacity conductor	\$4,784,000	100% (\$4,784,000)
CHINA-PONNEQUIN 100kV	Rebuild line with higher-capacity conductor	\$6,920,000	53% (\$3,663,000)
SORENSEN SUBSTATION-ELTOPIA 100kV L	Rebuild line with higher-capacity conductor	\$2,568,000	100% (\$2,568,000)
BARRE-HIGH SPRINGS 100kV	Rebuild line with higher-capacity conductor	\$3,754,000	100% (\$3,754,000)
DEER CRK-PONNEQUIN 100kV Line	Rebuild line with higher-capacity conductor	\$4,731,000	53% (\$2,505,000)
WAYNEDEALE-INDEGO 100kV	Rebuild line with higher-capacity conductor	\$2,330,000	36% (\$834,000)
BARRE-STAGECOACH 100kV	Rebuild line with higher-capacity conductor	\$9,438,000	100% (\$9,438,000)
WAYNEDEALE SUBSTATION-AEP PETTIT AVENUE 100kV #1	Rebuild line with higher-capacity conductor	\$1,265,000	36% (\$451,000)
WAYNEDEALE-AEP PETTIT AVENUE 100kV	Rebuild line with higher-capacity conductor	\$1,922,000	36% (\$685,000)
INDEGO-COLUMBIA HILLS 100kV	Rebuild line with higher-capacity conductor	\$1,998,000	70% (\$1,397,000)
STAUNTON-PIQUA POWER PLANT 230kV Line	Rebuild line with higher-capacity conductor	\$16,998,000	5% (\$835,000)
ILLINOIS ROAD-HEBRON ROSEMEADE 100kV	Rebuild line with higher-capacity conductor	\$1,598,000	50% (\$804,000)
SEGS 5-AEP PETTIT AVENUE 100kV	Rebuild line with higher-capacity conductor	\$1,656,000	36% (\$600,000)
CARLEY ROAD-PIQUA POWER PLANT 230kV	Rebuild line with higher-capacity conductor	\$35,424,000	5% (\$1,740,000)
STAUNTON-BARAGA VLG 230kV	Rebuild line with higher-capacity conductor	\$46,492,000	5% (\$2,284,000)
	Total allocated to project		\$86,368,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.